

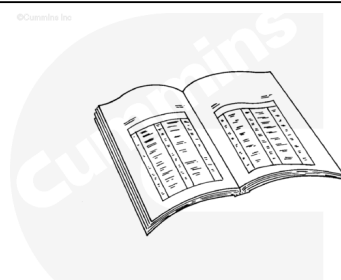
Trainings ▾

Preparatory Steps

Automotive with CM870

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html>)
- Remove the variable geometry turbocharger (VGT). Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)

- Remove the exhaust gas recirculation (EGR) valve. Refer to Procedure 011-022 in Section 11.

(/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)

- Remove the EGR cooler connection. Refer to Procedure 011-024 in Section 11.

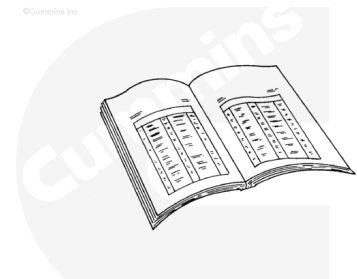
(/qs3/pubsys2/xml/en/procedures/10/10-011-024.html)

- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)

Automotive With CM871

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)

- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html)
- Remove the VGT. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Remove the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)
- Remove the EGR cooler connection. Refer to Procedure 011-024 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-024.html)
- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)

Automotive, Industrial with CM570 and Power Generation without CM570

WARNING

©Cummins Inc.

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

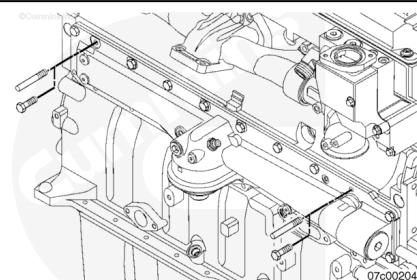
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html)
- Remove the turbocharger. This step is **not** required with a die cast oil cooler. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html)

Remove

All Applications

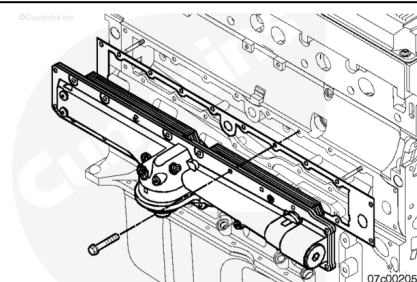
Clean the area above the oil cooler to prevent debris entry into the oil passages.

Remove two of the lubricating oil cooler mounting capscrews and install two guide pins.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

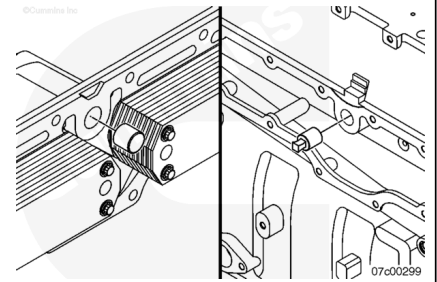


Remove the remaining 18 capscrews, oil cooler, and gasket. Discard the gasket.

⚠ CAUTION ⚠

Immediately upon removal of the lubricating oil cooler assembly, a plug must be inserted into the housing and cylinder block oil passage drillings. Failure to insert the oil

passage plug can result in a bearing failure, crankshaft failure, or both.



The reusable plugs, Part Numbers 4918322, 4918323, and 4918340, are included in Protective Plug Kit, Part Number 4918319. These plugs are necessary to prevent debris from entering the lubrication system during the repair.

Plug the oil cooler housing filtered oil passage with plug, Part Number 4918323, the block filtered oil passage with plug, Part Number 4918340, and the block oil pump passage with plug, Part Number 4918322.

Firmly push the plugs into the oil passages to prevent all debris from entering the lubrication system when cleaning the gasket surface.

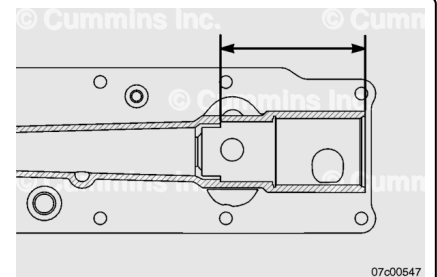
Clean and Inspect for Reuse

All Applications

Inspect the oil thermostat seat depth.

Remove the oil thermostat. Refer to Procedure 007-039 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-039.html>)

Measure from the spot face to the oil thermostat seat. Reference graphic for measurement points. If the measured dimension for lubricating oil cooler housing with integrated filter head is less than the minimum dimension, the oil cooler housing will need to be replaced.

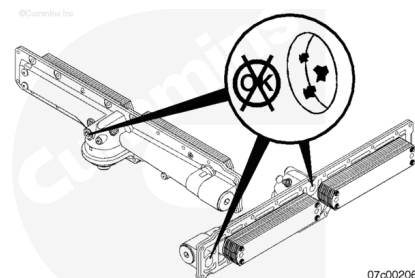


Lubricating Oil Cooler Housing Seat Depth		
mm		in
123.47	MIN	4.861

Install the oil thermostat. Refer to Procedure 007-039 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-039.html>)

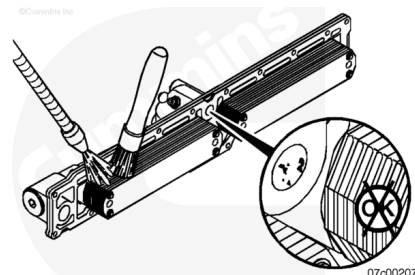
⚠ CAUTION ⚠

If the oil passages of the lubricating oil cooler become contaminated with metallic debris or dirt, the lubricating oil cooler must be replaced. Contamination can be defined by metal in the lubricating oil filter, and/or by debris damage to the bearings, or debris damage of another component, such that they do not meet reuse guidelines.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

If the oil passages of the lubricating oil cooler become contaminated with metallic debris or dirt, the lubricating oil cooler must be replaced. Contamination can be defined by metal in the lubricating oil filter, and/or by debris damage to the bearings, or debris damage of another component, such that they do not meet reuse guidelines.

Plug the oil cooler passages with rubber plugs, Part Numbers 4918322, 4918323; and 4918340; included in kit, Part Number 4918319, to prevent debris from entering the cooler when cleaning the gasket surface.

Clean the gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258. If the gasket material residue can **not** be felt with a finger, the surface is ready to accept the new gasket.

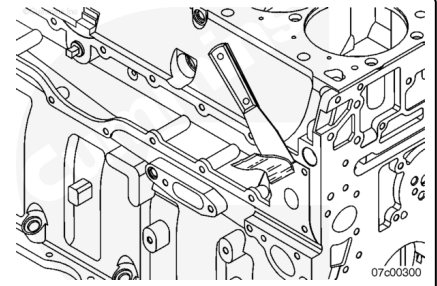
Use solvent to clean and flush the oil cooler passages and cores.

Blow dry with compressed air.

Remove the rubber plugs when the cleaning is complete.

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cylinder block when cleaning the gasket sealing surfaces. Connecting rod bearing failures can be caused if debris is introduced into the cylinder block or lubricating oil cooler housing oil passages. Therefore, use of power tools combined with abrasive pads to clean gasket surfaces is not recommended.



Plug the oil passages in the cylinder block with rubber plugs.

Clean the oil cooler to cylinder block gasket surface with Scotch-Brite™ 7448 abrasive pad.

Remove the rubber plugs when the cleaning is complete.

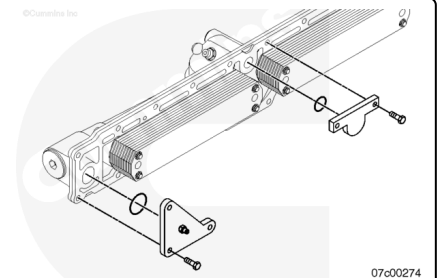
Leak Test

All Applications

It is important to leak test the oil cooler assembly as a unit, so leaks at the mounting joints can be identified. These leaks can **not** be found when testing individual elements that have been removed from the assembly. An element is replaced **only** if it is found to be leaking.

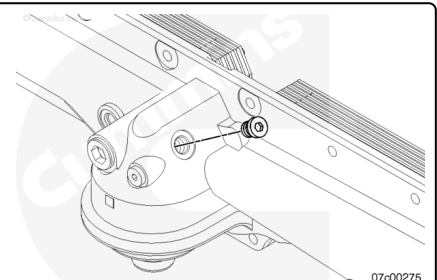
This procedure is **most** effective when the oil cooler assembly is tested in water below 7°C [44°F].

Use the mounting capscrews from the lubricating oil cooler to attach the two plates and corresponding o-rings from Oil Cooler Leak Test Kit, Part Number 4918200, to the lubricating oil cooler housing.



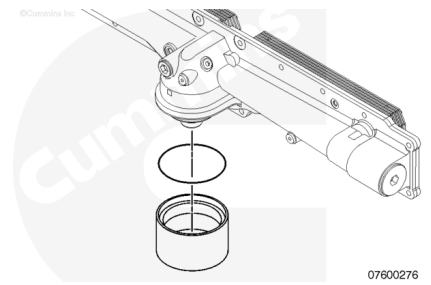
Remove the threaded male union that connects to the turbocharger oil supply.

Install the threaded plug, Part Number 3089567.



Insert the o-ring from the oil filter into the filter head cap, Part Number 4918204.

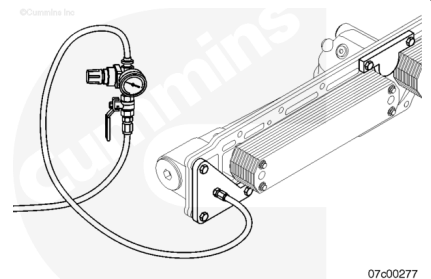
Use engine oil to lubricate the gasket surface and install the filter head cap.



Attach air pressure regulator, Part Number 3164231, to the outlet plate air fitting and pressurize the assembly to 137 kPa [20 psi].

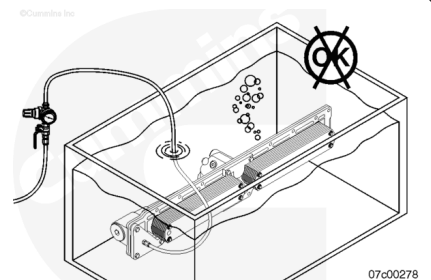
Pressure decay readings **must** be observed three consecutive times to be considered accurate.

Monitor pressure gauge and determine rate of pressure decay for 15 minutes. Allowable pressure drop is 7kPa [1 psi] maximum.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

If the lubricating oil cooler does **not** meet the maximum pressure decay requirements from the leak test, perform the following dunk test to identify the probable source of the leak

Place the oil cooler assembly under water.

If bubbles are observed, carefully determine the source of the bubbles and replace the components as necessary.

If the element is suspected of leaking, use the following procedure for leak testing individual elements. Refer to Procedure 007-007 in Section 7.

(/qs3/pubsys2/xml/en/procedures/10/10-007-007.html)

Mounting joint leaks are much more common than element leaks. Do **not** replace components that do **not** leak.

If no leaks are found and the elements are to be reused, replace the capscrews and gaskets if the engine serial number is in the range of 14023600 to 1405000 or 7900000 to

79012457.

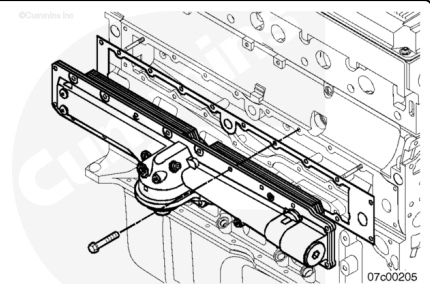
If no leaks are found and the engine serial number is **not** in the above ranges, the element does **not** get removed, but the capscrew torque is to be checked and then tightened to the specified torque.

Install

All Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



When installing a new two-element oil cooler on an older engine, be sure the cooler element mounting capscrews do **not** interfere with the cylinder block.

Install the guide pins.

Remove the protective plugs from the oil passages. Make sure no debris enters the lubricating system. Use a clean shop towel to carefully wipe out the oil passages and inspect to make sure no debris is left in the passages.

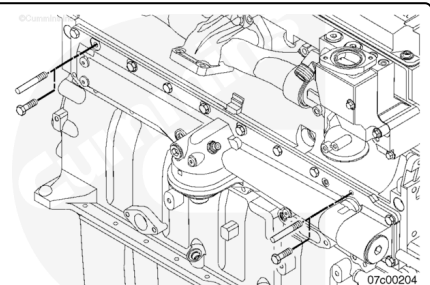
Install a new mounting gasket and the lubricating oil cooler housing assembly.

Install the 18 capscrews.

Remove the guide pins and install the two remaining capscrews.

Tighten the capscrews in a circular pattern from the center out.

Torque Value: 47 n•m [35 ft-lb]

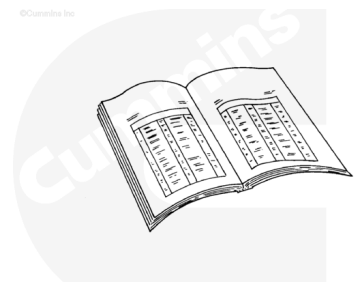


Finishing Steps

Automotive with CM870

- Install the EGR cooler. Refer to Procedure 011-019 in Section 19. (</qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html>)
- Install the EGR cooler connections. Refer to Procedure 011-024 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-024.html>)
- Install the EGR valve. Refer to Procedure 011-022 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html>)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html>)
- Install the VGT. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)
- Fill the engine with clean oil.
- Operate the engine to normal operating range. Check for oil and coolant leaks.

©Cummins Inc.

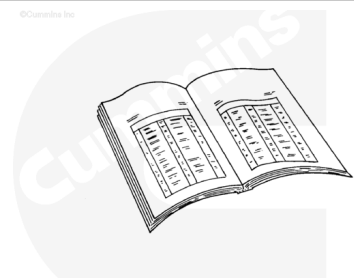


ck800wa

Automotive With CM871

- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html>)
- Install the VGT. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)

©Cummins Inc.



ck800wa

- Fill the engine with clean oil.
- Operate the engine to normal operating range. Check for oil and coolant leaks.

Automotive, Industrial with CM570 and Power Generation without CM570

- Install the oil filter. Install the alternator. Refer to Procedure 013-001 in Section 13.
(/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Fill the engine with clean oil.
- Operate the engine to normal operating range. Check for oil and coolant leaks.